VISUALIZATION TOOL FOR ELECTRIC VEHICLE CHARGE AND RANGE ANALYSIS

Internship Report

Submitted by

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Submitted in partial fulfillment of the requirements for the Internship Program

Last Line

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CERTIFICATE

This is to certify that **Mr. Manchala Karthik**, a student of **B.Tech in Artificial Intelligence and Data Science**, has successfully completed the internship project titled **"Visualization Tool for Electric Vehicle Charge and Range Analysis"**.

During the internship, the student worked on analyzing electric vehicle data using **MySQL** for database management and **Tableau** for data visualization. The project involved data collection, database creation, SQL query execution, dashboard development, and visualization of electric vehicle charging and range analysis. The student completed the assigned work sincerely and demonstrated good technical knowledge, dedication, and enthusiasm throughout the internship period.

This report is submitted in partial fulfillment of the internship requirements.

Faculty Guide　　　　　　　　　　　　　　　　　　Head of Department

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the successful completion of my internship project titled **"Visualization Tool for Electric Vehicle Charge and Range Analysis."**

I am thankful to my internship mentors for providing valuable guidance, encouragement, and technical knowledge during the entire internship. Their suggestions and support helped me understand the concepts of MySQL database management and Tableau data visualization.

I also express my heartfelt thanks to the faculty members of my college for their continuous motivation and support throughout the internship period.

Finally, I would like to thank my parents, friends, and everyone who directly or indirectly contributed to the successful completion of this project.

ABSTRACT

Electric Vehicles (EVs) are becoming increasingly important due to their environmental benefits and growing adoption worldwide. Analyzing EV data helps users understand vehicle performance, charging capabilities, battery capacity, pricing, and driving range.

The main objective of this project is to develop a visualization tool for Electric Vehicle Charge and Range Analysis using **MySQL** and **Tableau**. MySQL is used to store and manage the dataset, while Tableau is used to create interactive charts, dashboards, and stories for better understanding of the data.

The project includes database creation, SQL query execution, data preprocessing, visualization development, dashboard creation, and story presentation. Various visualizations such as EV range analysis, efficiency analysis, brand comparison, price analysis, and charging station analysis are developed to provide meaningful insights.

This project demonstrates how data visualization techniques can simplify complex information and support better decision-making. It also enhances practical knowledge of database management, SQL, and business intelligence tools.

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CHAPTER 1

INTRODUCTION

Electric Vehicles (EVs) are transforming the automobile industry by providing an environmentally friendly alternative to conventional fuel-powered vehicles. With increasing concerns about air pollution, climate change, and fuel consumption, governments and automobile manufacturers are encouraging the adoption of electric vehicles across the world.

As the number of electric vehicles continues to grow, a large amount of data related to battery capacity, charging time, driving range, vehicle price, and brand information is generated. Analyzing this data helps users understand vehicle performance, compare different EV models, and make informed decisions while purchasing electric vehicles.

This project, titled **"Visualization Tool for Electric Vehicle Charge and Range Analysis,"** focuses on analyzing electric vehicle data using **MySQL** and **Tableau**. MySQL is used to store and manage the dataset efficiently, while Tableau is used to create interactive visualizations, dashboards, and stories that present meaningful insights in an easy-to-understand format.

The project demonstrates how data visualization techniques can simplify complex datasets and help identify important trends related to electric vehicle performance, charging infrastructure, pricing, and driving range. The developed dashboard provides users with an effective way to analyze electric vehicle information and supports better decision-making through visual analytics.

CHAPTER 2

PROBLEM STATEMENT

The rapid growth of electric vehicles has resulted in the generation of a large amount of data related to battery capacity, charging time, vehicle range, pricing, and manufacturer details. Although this information is available, it is often stored in raw datasets that are difficult to understand and analyze.

Without proper visualization, comparing different electric vehicles and identifying useful patterns becomes time-consuming and inefficient. Users may find it difficult to evaluate vehicle performance, charging efficiency, and price differences using only tabular data.

The major challenge is to convert raw electric vehicle data into meaningful visual representations that help users quickly understand important insights. Therefore, there is a need for an efficient visualization system that stores the data in a structured database and presents it through interactive dashboards and charts.

This project addresses this problem by using MySQL for database management and Tableau for creating interactive visualizations that improve data analysis and support better decision-making.

CHAPTER 3

OBJECTIVES

The main objectives of this project are:

• To collect and organize electric vehicle data in a structured database.

• To create a MySQL database for efficient storage and management of EV information.

• To perform SQL queries for extracting useful insights from the dataset.

• To analyze electric vehicle data using Tableau.

• To create interactive charts, dashboards, and stories for better visualization.

• To compare electric vehicles based on driving range, battery capacity, charging time, price, and brand.

• To present meaningful insights that support better decision-making for users and stakeholders.

• To improve practical knowledge of database management and data visualization tools.

CHAPTER 4

SOFTWARE AND HARDWARE REQUIREMENTS

The successful implementation of the project required both software and hardware resources. The following requirements were used during the development of the project.

Software Requirements

Software	Purpose
MySQL Workbench	Database creation and SQL query execution
Tableau Desktop/Public	Data visualization, dashboards and stories
Microsoft Word	Report preparation
Google Chrome	Accessing online resources and GitHub

Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 or above
RAM	8 GB or above
Storage	256 GB SSD / 500 GB HDD
Operating System	Windows 10/11
Internet Connection	Required for GitHub and online resources

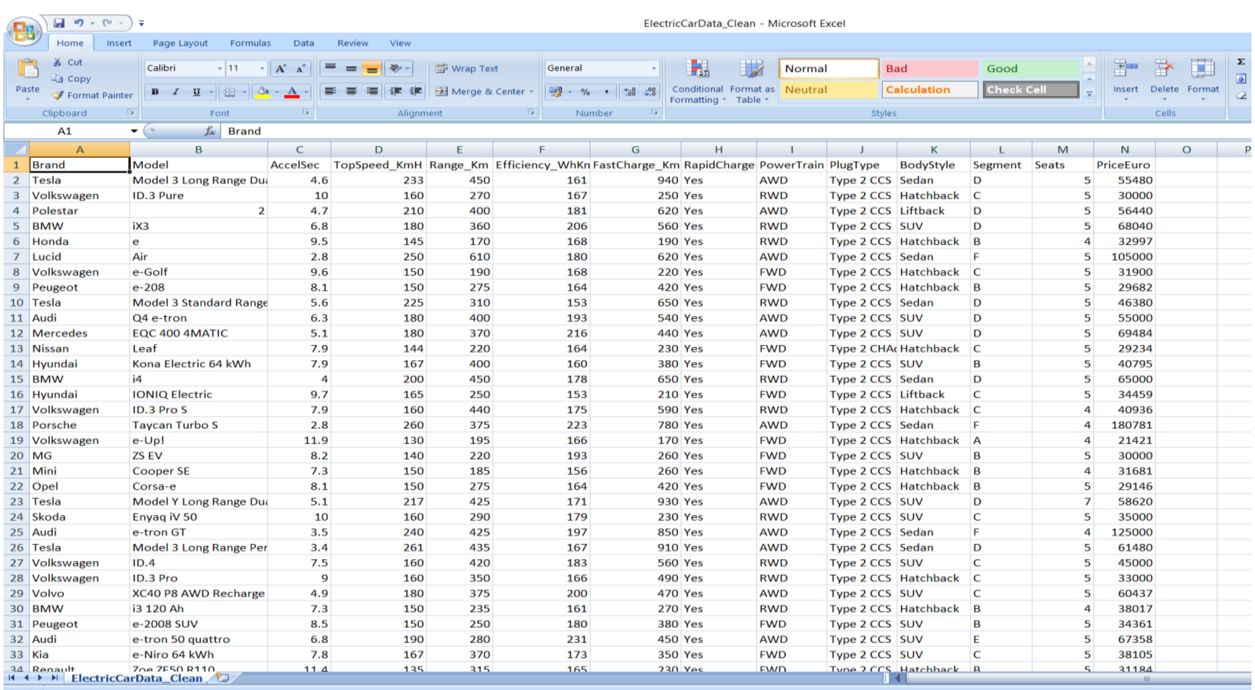
CHAPTER 5

DATASET DESCRIPTION

The dataset used in this project contains information about various electric vehicles and their technical specifications. It was collected to analyze important parameters such as battery capacity, charging time, driving range, vehicle price, efficiency, and manufacturer details.

The dataset was imported into MySQL for efficient storage and management. After importing the data, SQL queries were executed to retrieve useful information required for analysis. The processed data was then connected to Tableau to create interactive visualizations, dashboards, and stories.

The dataset forms the foundation of this project by providing the necessary information required for meaningful analysis and visualization of electric vehicle performance.



	Brand	Model	AccelSec	TopSpeed_KmH	Range_Km	Efficiency_WhKn	FastCharge_Km	RapidCharge	PowerTrain	PlugType	BodyStyle	Segment	Seats	PriceEuro
2	Tesla	Model 3 Long Range Dua	4.6	233	450	161	940	Yes	AWD	Type 2 CCS	Sedan	D	5	55480
3	Volkswagen	ID.3 Pure	10	160	270	167	250	Yes	RWD	Type 2 CCS	Hatchback	C	5	30000
4	Polestar	2	4.7	210	400	181	620	Yes	AWD	Type 2 CCS	Liftback	D	5	56440
5	BMW	iX3	6.8	180	360	206	560	Yes	RWD	Type 2 CCS	SUV	D	5	68040
6	Honda	e	9.5	145	170	168	190	Yes	RWD	Type 2 CCS	Hatchback	B	4	32997
7	Lucid	Air	2.8	250	610	180	620	Yes	AWD	Type 2 CCS	Sedan	F	5	105000
8	Volkswagen	e-Golf	9.6	150	190	168	220	Yes	FWD	Type 2 CCS	Hatchback	C	5	31900
9	Peugeot	e-208	8.1	150	275	164	420	Yes	FWD	Type 2 CCS	Hatchback	B	5	29682
10	Tesla	Model 3 Standard Range	5.6	225	310	153	650	Yes	RWD	Type 2 CCS	Sedan	D	5	46380
11	Audi	Q4 e-tron	6.3	180	400	193	540	Yes	AWD	Type 2 CCS	SUV	D	5	55000
12	Mercedes	EQC 400 4MATIC	5.1	180	370	216	440	Yes	AWD	Type 2 CCS	SUV	D	5	69484
13	Nissan	Leaf	7.9	144	220	164	230	Yes	FWD	Type 2 CHA(	Hatchback	C	5	29234
14	Hyundai	Kona Electric 64 kWh	7.9	167	400	160	380	Yes	FWD	Type 2 CCS	SUV	B	5	40795
15	BMW	i4	4	200	450	178	650	Yes	RWD	Type 2 CCS	Sedan	D	5	65000
16	Hyundai	IONIQ Electric	9.7	165	250	153	210	Yes	FWD	Type 2 CCS	Liftback	C	5	34459
17	Volkswagen	ID.3 Pro S	7.9	160	440	175	590	Yes	RWD	Type 2 CCS	Hatchback	C	4	40936
18	Porsche	Taycan Turbo S	2.8	260	375	223	780	Yes	AWD	Type 2 CCS	Sedan	F	4	180781
19	Volkswagen	e-Up!	11.9	130	195	166	170	Yes	FWD	Type 2 CCS	Hatchback	A	4	21421
20	MG	ZS EV	8.2	140	220	193	260	Yes	FWD	Type 2 CCS	SUV	B	5	30000
21	Mini	Cooper SE	7.3	150	185	156	260	Yes	FWD	Type 2 CCS	Hatchback	B	4	31681
22	Opel	Corsa-e	8.1	150	275	164	420	Yes	FWD	Type 2 CCS	Hatchback	B	5	29146
23	Tesla	Model Y Long Range Dua	5.1	217	425	171	930	Yes	AWD	Type 2 CCS	SUV	D	7	58620
24	Skoda	Enyaq iV 50	10	160	290	179	230	Yes	RWD	Type 2 CCS	SUV	C	5	35000
25	Audi	e-tron GT	3.5	240	425	197	850	Yes	AWD	Type 2 CCS	Sedan	F	4	125000
26	Tesla	Model 3 Long Range Per	3.4	261	435	167	910	Yes	AWD	Type 2 CCS	Sedan	D	5	61480
27	Volkswagen	ID.4	7.5	160	420	183	560	Yes	RWD	Type 2 CCS	SUV	C	5	45000
28	Volkswagen	ID.3 Pro	9	160	350	166	490	Yes	RWD	Type 2 CCS	Hatchback	C	5	33000
29	Volvo	XC40 P8 AWD Recharge	4.9	180	375	200	470	Yes	AWD	Type 2 CCS	SUV	C	5	60437
30	BMW	i3 120 Ah	7.3	150	235	161	270	Yes	RWD	Type 2 CCS	Hatchback	B	4	38017
31	Peugeot	e-2008 SUV	8.5	150	250	180	380	Yes	FWD	Type 2 CCS	SUV	B	5	34361
32	Audi	e-tron 50 quattro	6.8	190	280	231	450	Yes	AWD	Type 2 CCS	SUV	E	5	67358
33	Kia	e-Niro 64 kWh	7.8	167	370	173	350	Yes	FWD	Type 2 CCS	SUV	C	5	38105
34	Renault	Zoe ZE50 R110	11.4	135	315	165	230	Yes	FWD	Type 2 CCS	Hatchback	B	5	31184

Figure 5.1: Electric Vehicle Dataset Preview

CHAPTER 6

DATABASE DESIGN AND MYSQL IMPLEMENTATION

MySQL was used as the database management system for storing and organizing the electric vehicle dataset. A dedicated database was created to maintain the data in a structured format, making it easier to retrieve and analyze information.

After creating the database, the dataset was imported into MySQL. Tables were created with appropriate fields to store details such as vehicle name, manufacturer, battery capacity, charging time, driving range, vehicle price, and other related attributes.

Using MySQL provided efficient data storage, faster query execution, and better organization of the electric vehicle information. This database served as the primary data source for Tableau visualizations developed in the later stages of the project.

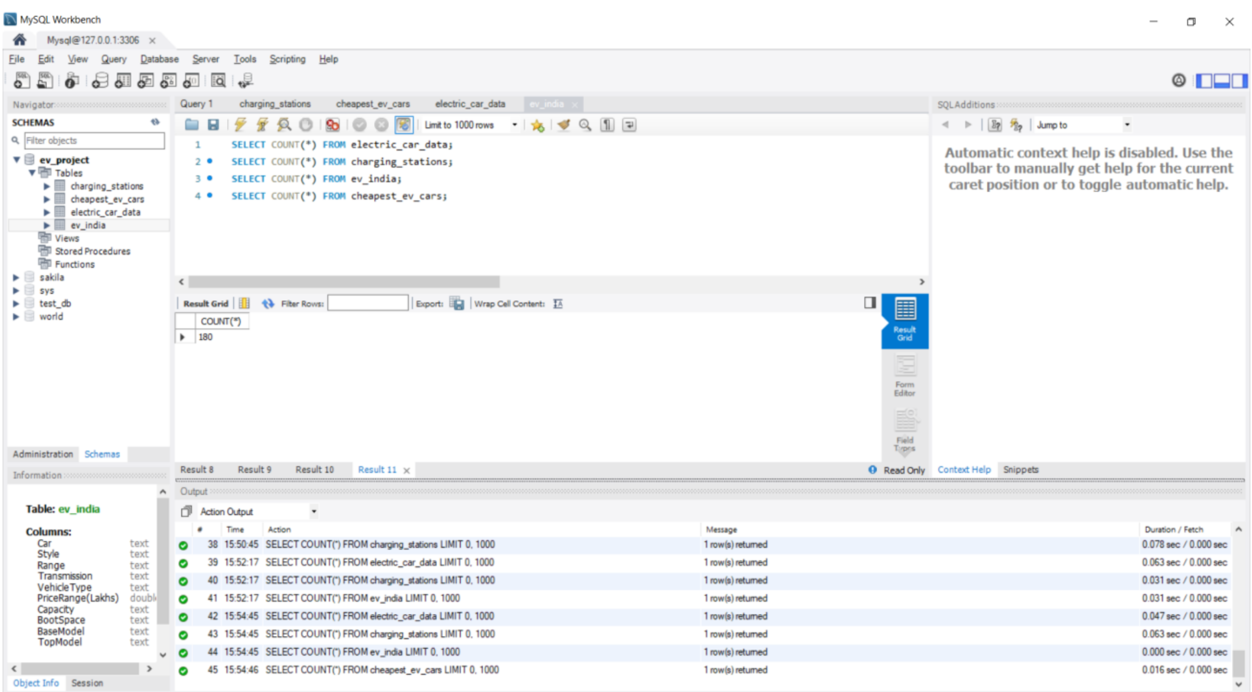


Figure 6.1: MySQL Database Creation

CHAPTER 7

TABLE CREATION AND DATA IMPORT

After creating the database, a table was created to store the electric vehicle dataset. The table structure was designed with appropriate columns to maintain all the required vehicle information, including vehicle name, manufacturer, battery capacity, charging time, driving range, price, and other important attributes.

The electric vehicle dataset was then imported into the created table using MySQL Workbench. Importing the dataset into the database ensured that all records were stored in a structured manner and could be accessed efficiently through SQL queries.

The successful creation of the table and import of the dataset formed the foundation for further data analysis and visualization using Tableau.

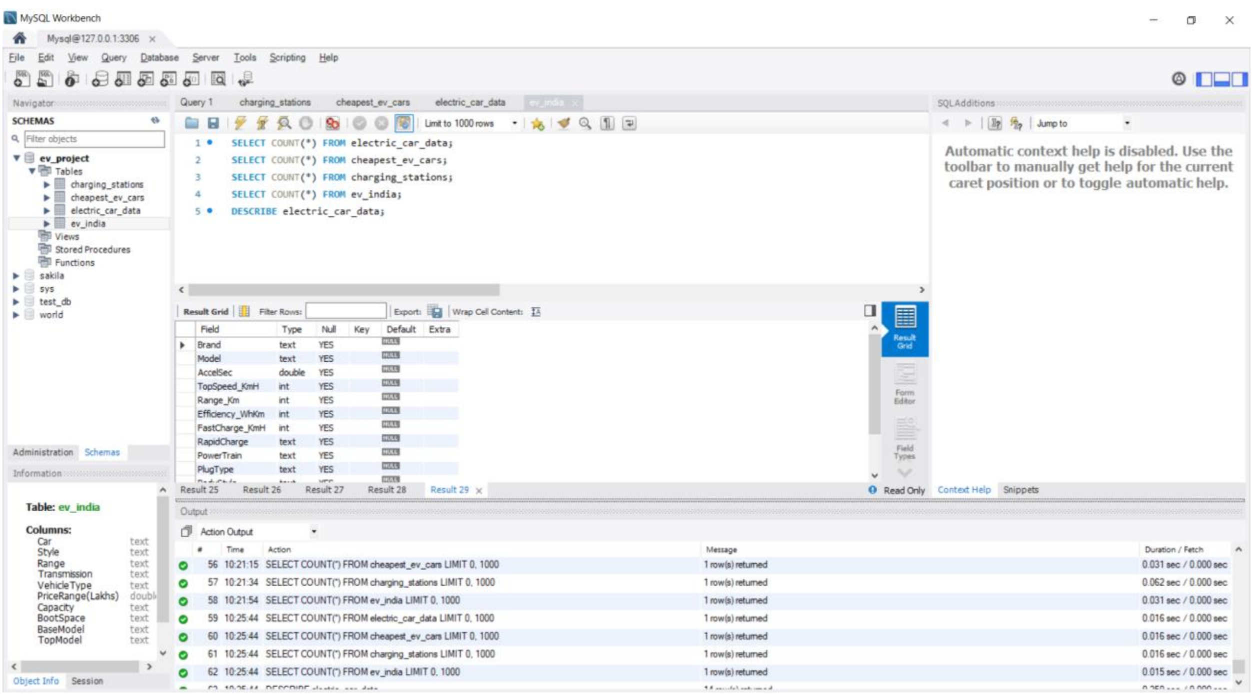


Figure7.1: Table Creation in MySQL

CHAPTER 8

SQL QUERY EXECUTION

After importing the dataset into MySQL, various SQL queries were executed to retrieve meaningful information from the database. SQL queries were used to filter records, display required information, sort data, and perform analytical operations.

These queries helped in understanding the electric vehicle dataset by extracting useful details such as vehicle specifications, charging information, driving range, pricing, and manufacturer details. The query results were further used for creating interactive visualizations in Tableau.

The execution of SQL queries ensured accurate data retrieval and improved the efficiency of data analysis before developing dashboards and stories.

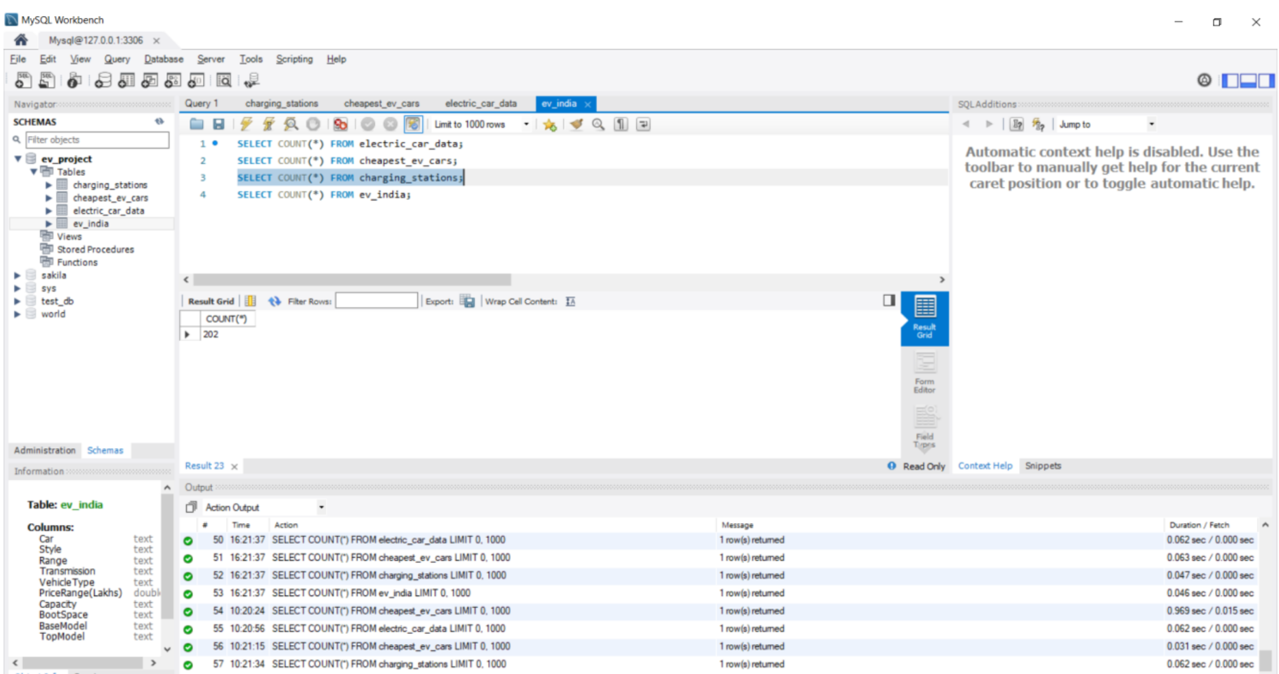


Figure 8.1: SQL Query Execution in MySQL Workbench

CHAPTER 9

DATA VALIDATION USING SQL

After importing the electric vehicle dataset into MySQL, SQL queries were executed to verify that the data had been imported correctly. One of the first validation steps was to calculate the total number of records available in the table.

The COUNT() function was used to determine the total number of entries stored in the database. This verification confirmed that the dataset had been imported successfully without any missing records.

Performing data validation before analysis improves the reliability of the project and ensures that the visualizations created in Tableau are based on complete and accurate information.

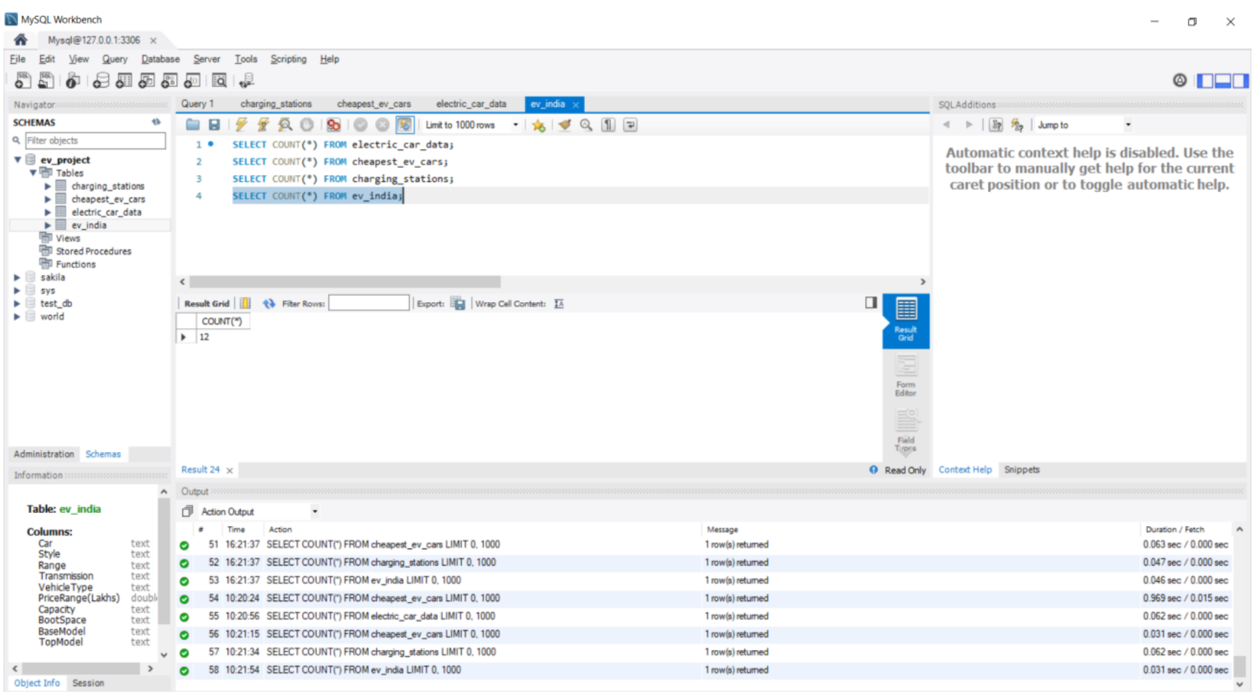


Figure 9.1: SQL Query Showing the Total Number of Records in the Electric Vehicle Table

CHAPTER 10

ELECTRIC CAR DATA TABLE STRUCTURE

The structure of the **Electric Car Data** table was examined using the **DESCRIBE** command in MySQL. This command displays detailed information about each column in the table, including the field name, data type, null values, key constraints, default values, and additional properties.

Understanding the table structure is an important step before performing data analysis because it ensures that the dataset has been imported correctly and that each attribute has the appropriate data type. It also helps in writing accurate SQL queries for retrieving and analyzing the required information.

The Electric Car Data table contains all the primary information related to electric vehicles, which serves as the main source of data for further analysis and visualization in Tableau.

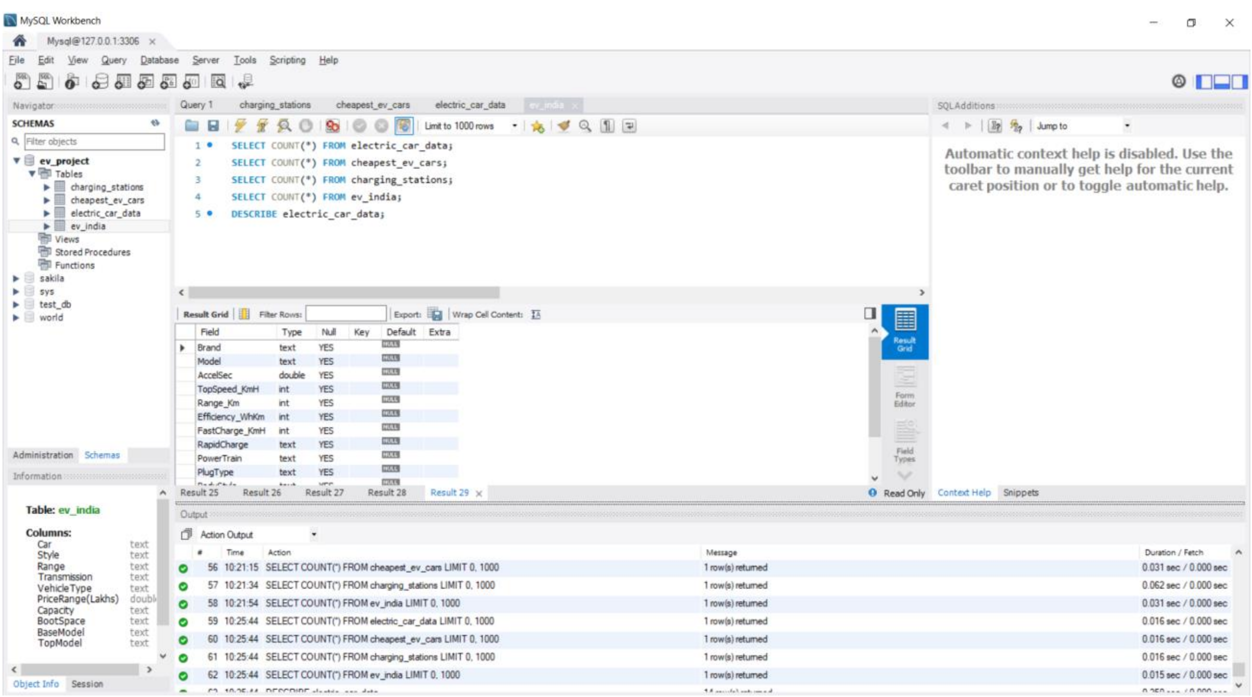


Figure 10.1: Structure of the Electric Car Data Table Using the DESCRIBE Command

CHAPTER 11

CHEAPEST EV CARS TABLE STRUCTURE

The **Cheapest EV Cars** table was analyzed using the **DESCRIBE** command in MySQL to understand its structure and verify the imported data. This command displays information about each column, including the field name, data type, null values, key information, default values, and other properties.

The table contains details of electric vehicles that are categorized based on their affordability. It provides useful information such as vehicle name, manufacturer, battery specifications, driving range, charging details, and price. This structured data helps in comparing affordable electric vehicles and identifying cost-effective options.

Examining the table structure before analysis ensures that the data is correctly organized and ready for visualization in Tableau.

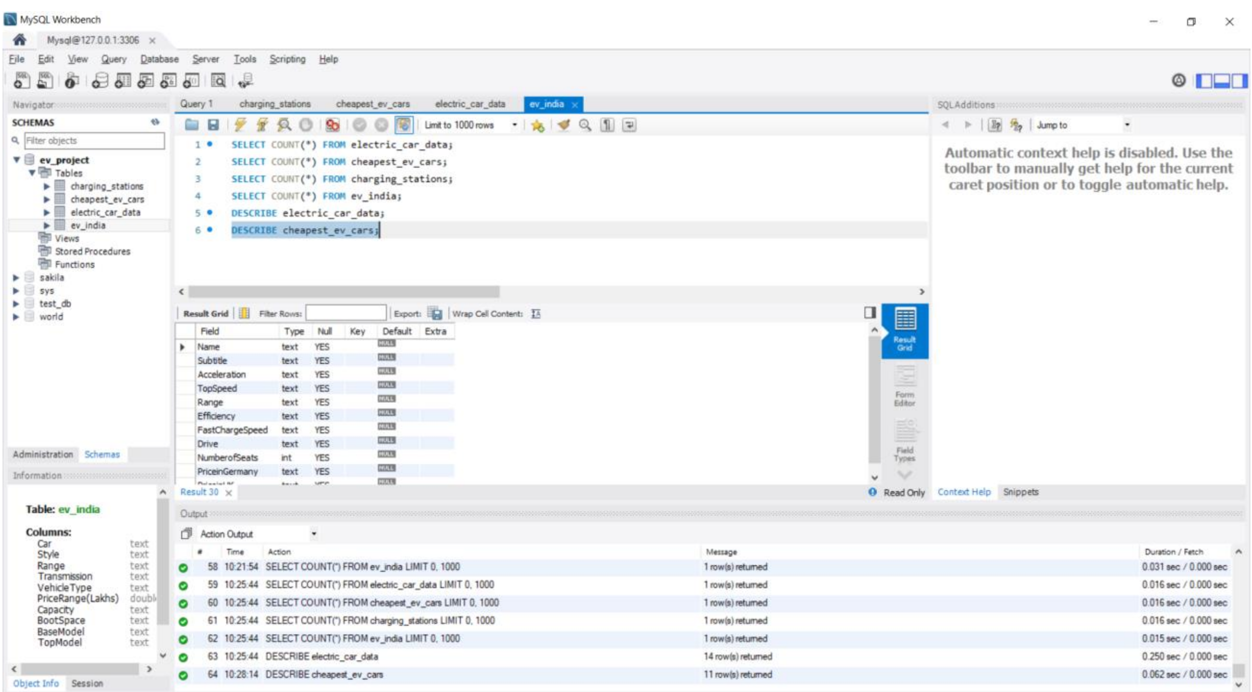


Figure 11.1: Structure of the Cheapest EV Cars Table Using the DESCRIBE Command

CHAPTER 12

CHARGING STATIONS TABLE STRUCTURE

The **Charging Stations** table was examined using the **DESCRIBE** command in MySQL to verify its structure and ensure that all required fields were correctly defined. This command displays the column names, data types, key constraints, null values, default values, and additional properties of the table.

The Charging Stations table contains information related to electric vehicle charging infrastructure. Properly organizing this information helps in analyzing charging station availability and supports the development of visualizations that provide meaningful insights into charging facilities.

Verifying the table structure before analysis ensures that the stored data is consistent, complete, and suitable for further processing and visualization using Tableau.

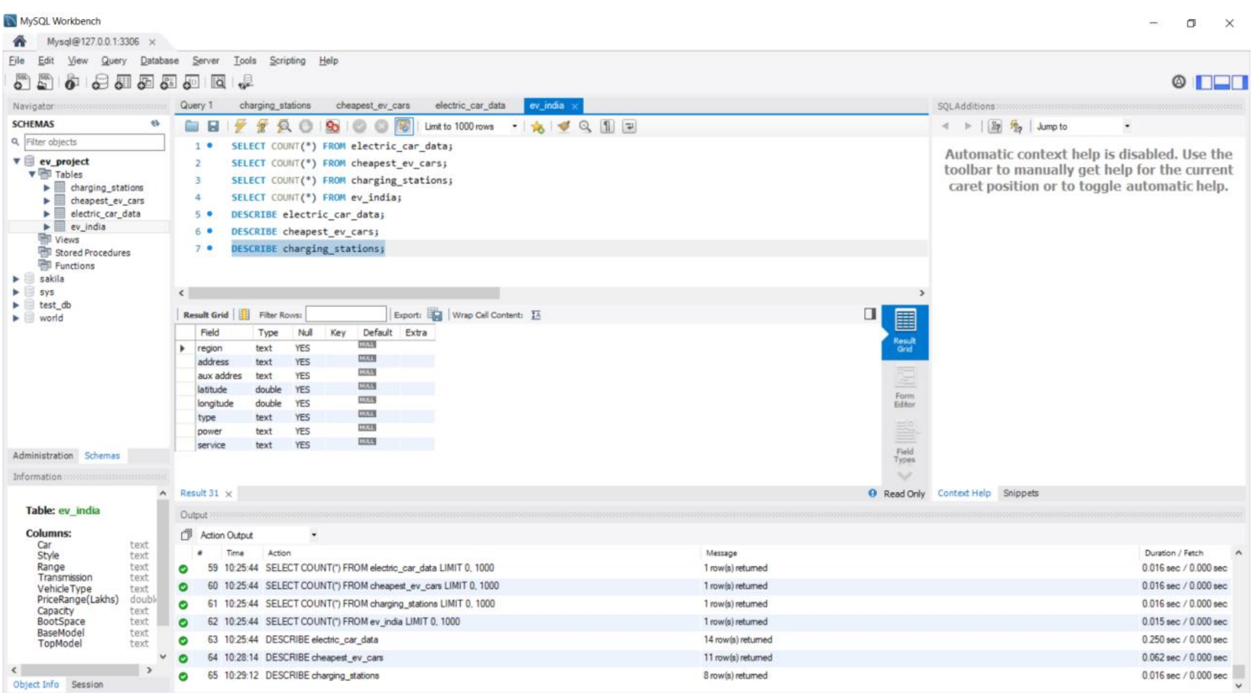


Figure 12.1: Structure of the Charging Stations Table Using the DESCRIBE Command

CHAPTER 13

EV INDIA TABLE STRUCTURE

The **EV India** table was examined using the **DESCRIBE** command in MySQL to verify its structure and ensure that the imported data was organized correctly. The DESCRIBE command displays detailed information about every column, including the field name, data type, null values, key constraints, default values, and additional properties.

The EV India table contains information related to electric vehicles available in the Indian market. It includes important attributes such as vehicle name, manufacturer, battery specifications, driving range, charging details, and pricing information. This data is useful for comparing electric vehicles available in India and identifying market trends.

Validating the table structure before performing analysis helps maintain data accuracy and provides a reliable foundation for creating visualizations and dashboards in Tableau.

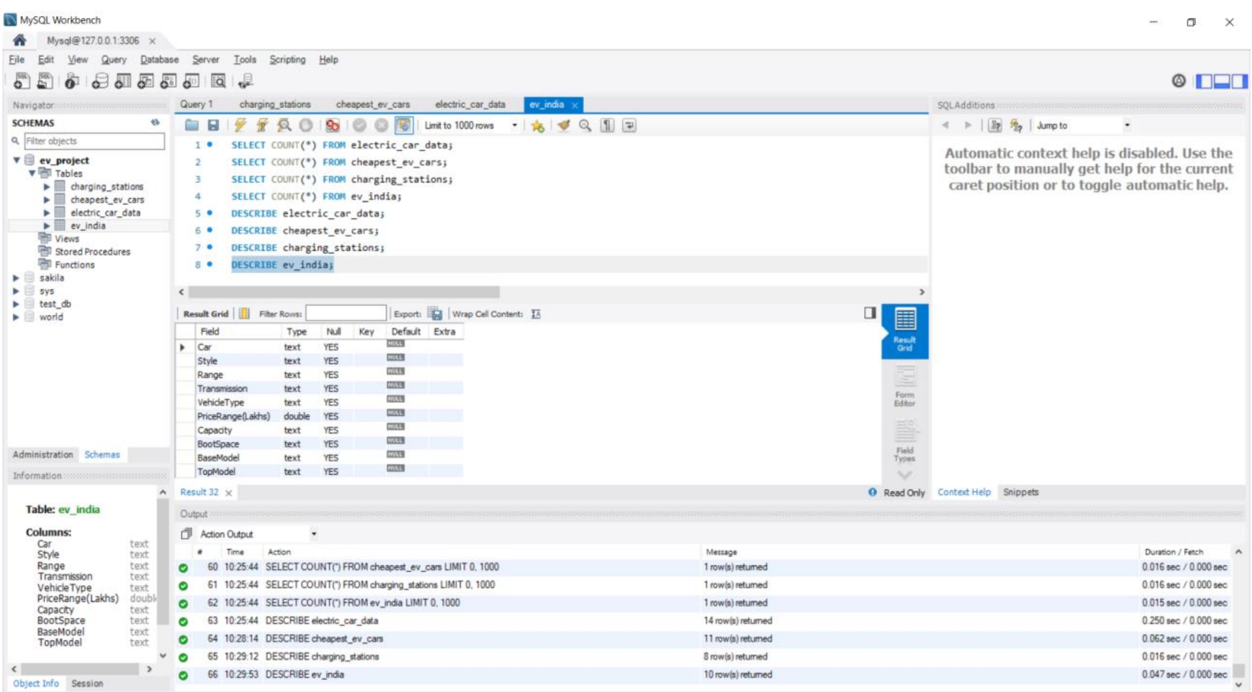


Figure 13.1: Structure of the EV India Table Using the DESCRIBE Command

CHAPTER 14

EV RANGE ANALYSIS USING TABLEAU

After completing the database implementation in MySQL, the processed data was connected to Tableau for visualization. Tableau was used to create interactive charts and dashboards that provide meaningful insights into the electric vehicle dataset.

The **EV Range Analysis** worksheet was developed to compare the driving range of different electric vehicles. This visualization enables users to identify vehicles with higher and lower driving ranges, making it easier to evaluate their performance and suitability for different travel requirements.

The graphical representation improves data interpretation by presenting complex information in a simple and visually appealing manner. It also helps users compare multiple vehicles efficiently without manually analyzing the dataset.

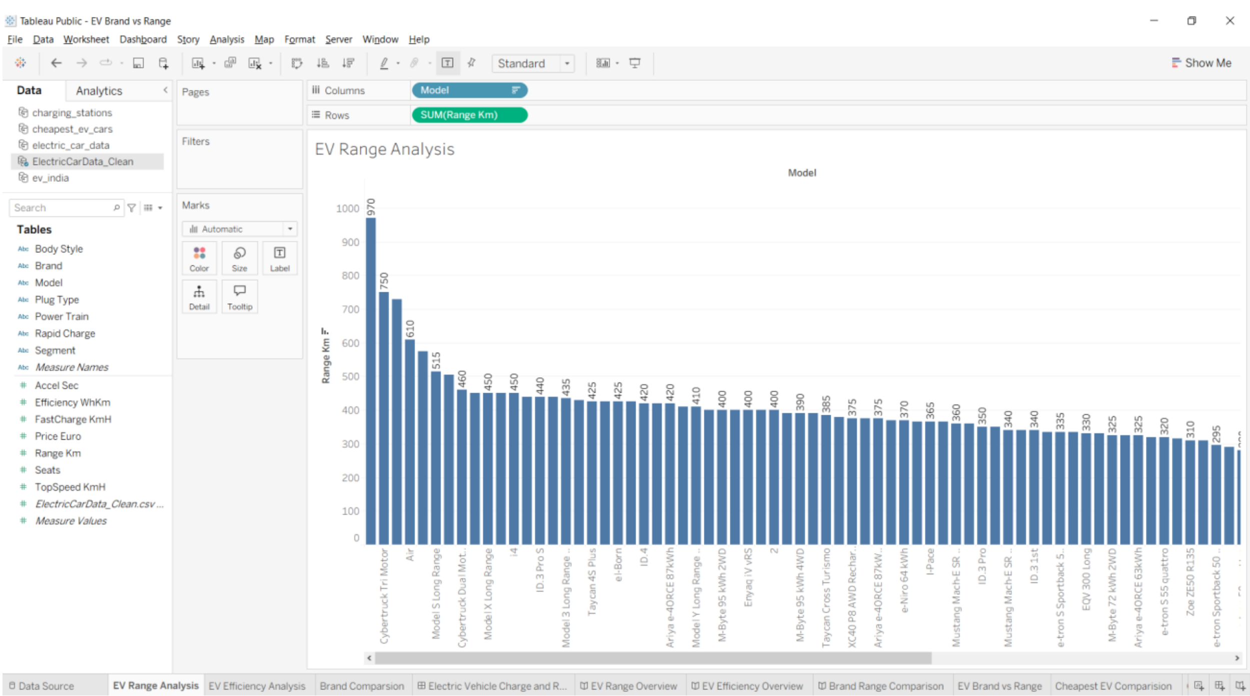


Figure 14.1: EV Range Analysis Worksheet in Tableau

CHAPTER 15

EV EFFICIENCY ANALYSIS USING TABLEAU

The **EV Efficiency Analysis** worksheet was created in Tableau to analyze and compare the efficiency of different electric vehicles. Vehicle efficiency is an important factor because it indicates how effectively an electric vehicle utilizes its battery power to achieve maximum driving range.

The visualization helps users compare the efficiency of multiple electric vehicles through an interactive graphical representation. It enables users to identify vehicles that provide better performance while consuming less energy.

Using Tableau's interactive features such as filtering, sorting, and visualization makes the comparison simple and improves the overall understanding of electric vehicle performance. This analysis assists users in selecting vehicles that offer better efficiency and value.

Figure 15.1: EV Efficiency Analysis Worksheet in Tableau

CHAPTER 16

BRAND COMPARISON USING TABLEAU

The **Brand Comparison** worksheet was developed in Tableau to compare different electric vehicle manufacturers based on the available dataset. This visualization enables users to understand how various brands perform with respect to the number of vehicle models and their overall presence in the dataset.

The comparison provides a clear overview of the manufacturers and helps identify brands that offer a wider range of electric vehicles. It also allows users to compare different companies quickly through graphical representation instead of manually reviewing the dataset.

Interactive features available in Tableau, such as filtering and highlighting, make the comparison more effective and user-friendly. This visualization helps users gain valuable insights into the electric vehicle market and the contribution of different manufacturers.

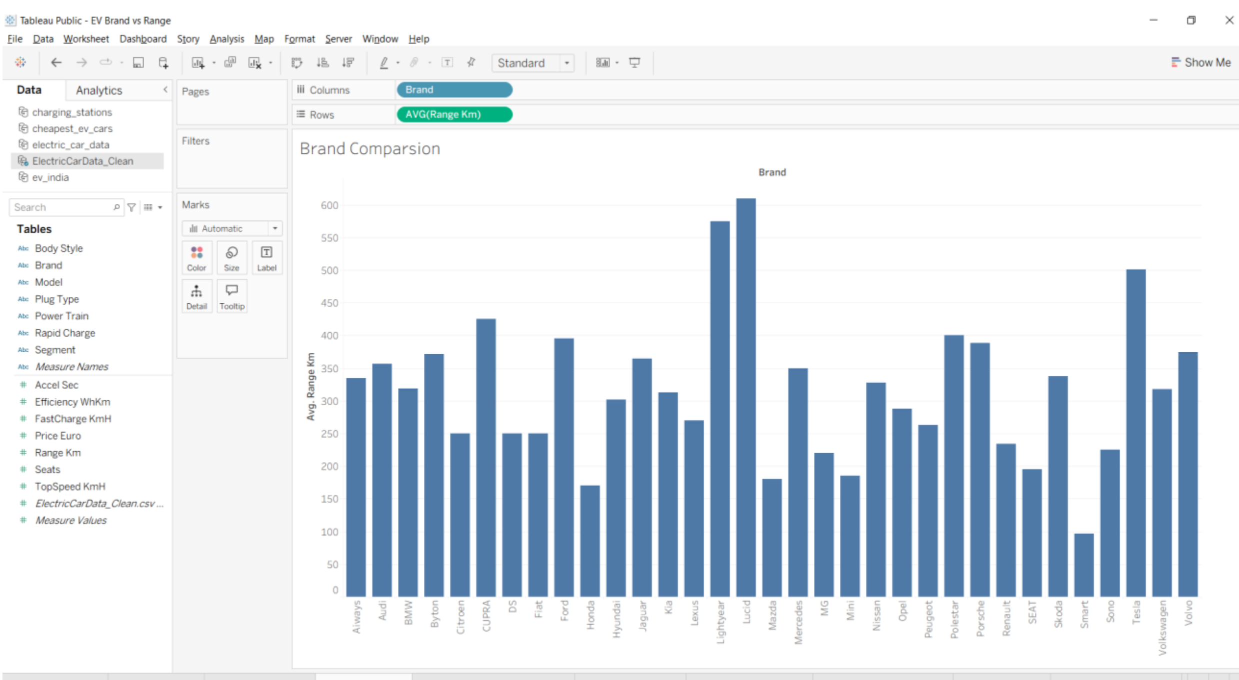


Figure 16.1: Brand Comparison Worksheet in Tableau

CHAPTER 17

EV BRAND VS RANGE ANALYSIS

The **EV Brand vs Range Analysis** worksheet was developed in Tableau to compare the driving range of electric vehicles across different manufacturers. Driving range is one of the most important factors considered by customers while selecting an electric vehicle.

This visualization helps identify which brands offer vehicles with higher driving ranges and enables users to compare the performance of multiple manufacturers in a single view. The graphical representation makes it easier to recognize trends and differences among brands.

By presenting the data visually, Tableau simplifies the comparison process and allows users to understand the strengths of different manufacturers. This analysis supports better decision-making for customers, researchers, and organizations interested in electric vehicle performance.

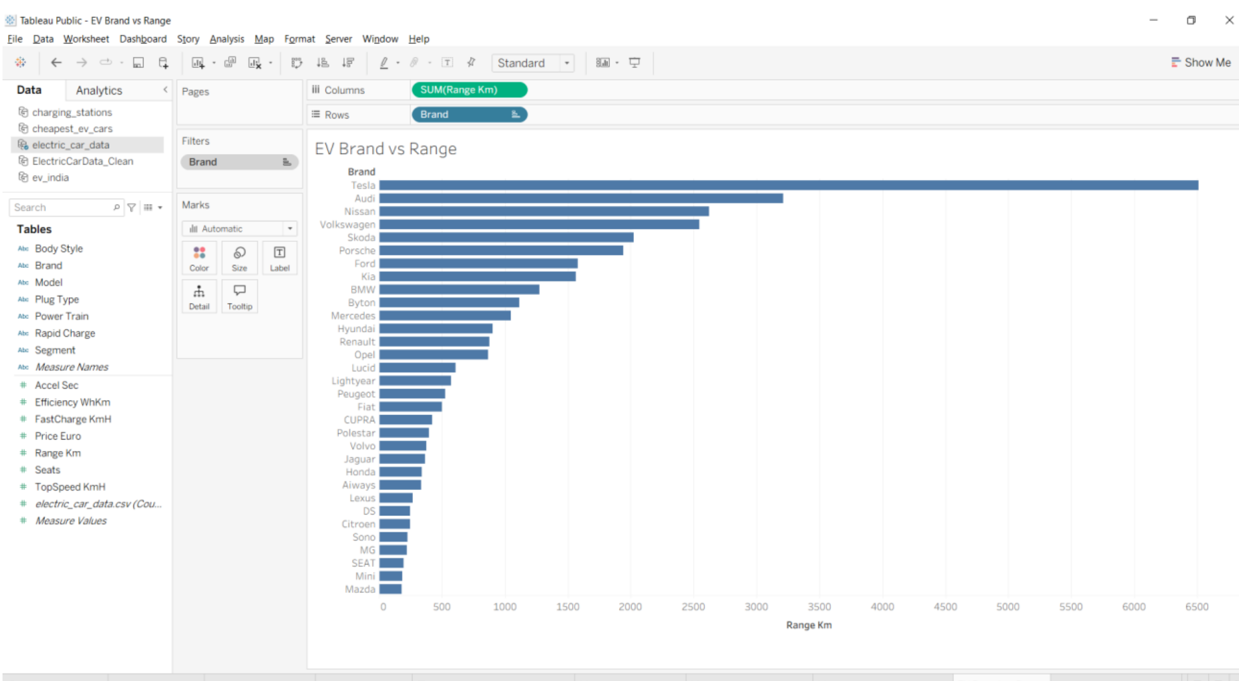


Figure 17.1: EV Brand vs Range Analysis Worksheet in Tableau

CHAPTER 18

CHEAPEST EV COMPARISON

The **Cheapest EV Comparison** worksheet was created in Tableau to compare electric vehicles based on their price. Price is one of the most important factors influencing the purchase decision of customers. This visualization helps identify affordable electric vehicles while comparing their basic features.

The comparison enables users to analyze different EV models and understand the relationship between vehicle cost and other specifications. By presenting the information graphically, users can easily recognize the most budget-friendly electric vehicles available in the dataset.

The visualization improves decision-making by allowing users to compare multiple vehicles quickly and effectively. Tableau's interactive features further enhance the analysis by providing filtering and comparison capabilities.

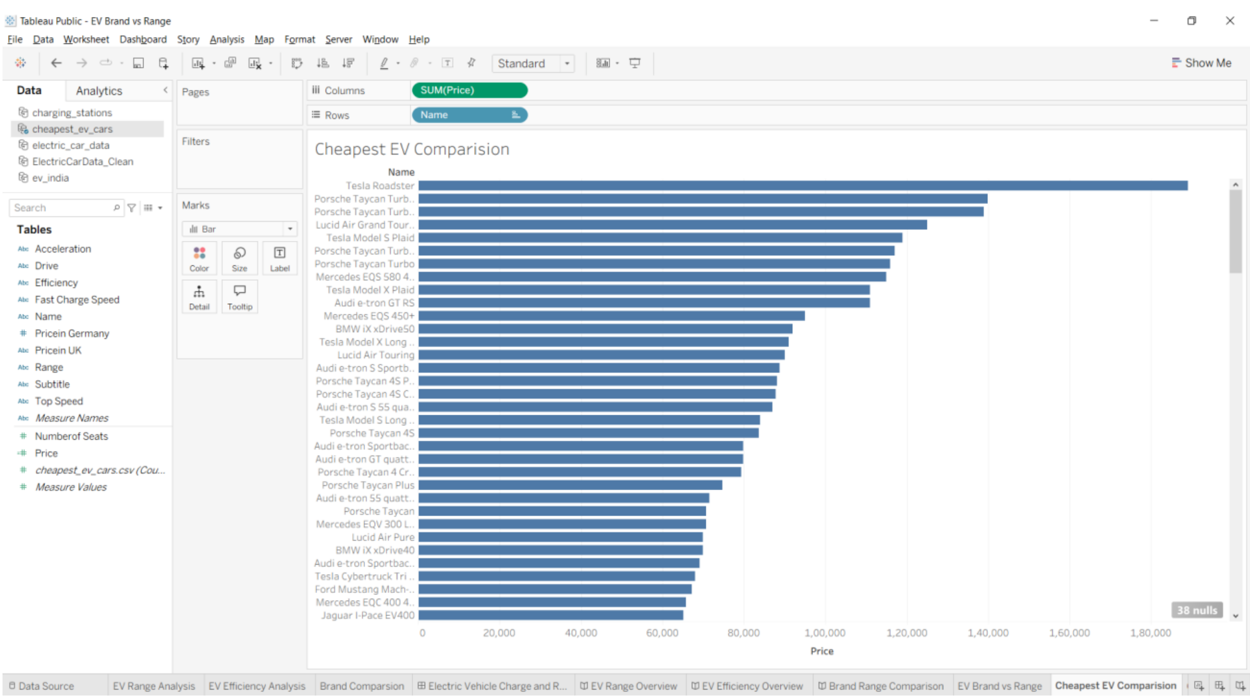


Figure 18.1: Cheapest EV Comparison Worksheet in Tableau

CHAPTER 19

CHARGING STATION ANALYSIS

The **Charging Station Analysis** worksheet was developed in Tableau to visualize the distribution and availability of electric vehicle charging stations. A reliable charging infrastructure plays a vital role in the successful adoption of electric vehicles, making this analysis an important part of the project.

The visualization provides a clear overview of charging station locations and helps users understand the availability of charging facilities. It enables comparisons between different locations and highlights areas where charging infrastructure is well established.

By using Tableau's interactive mapping and visualization features, the charging station data is presented in an easy-to-understand format. This analysis supports better planning for electric vehicle users and provides valuable insights into charging infrastructure.

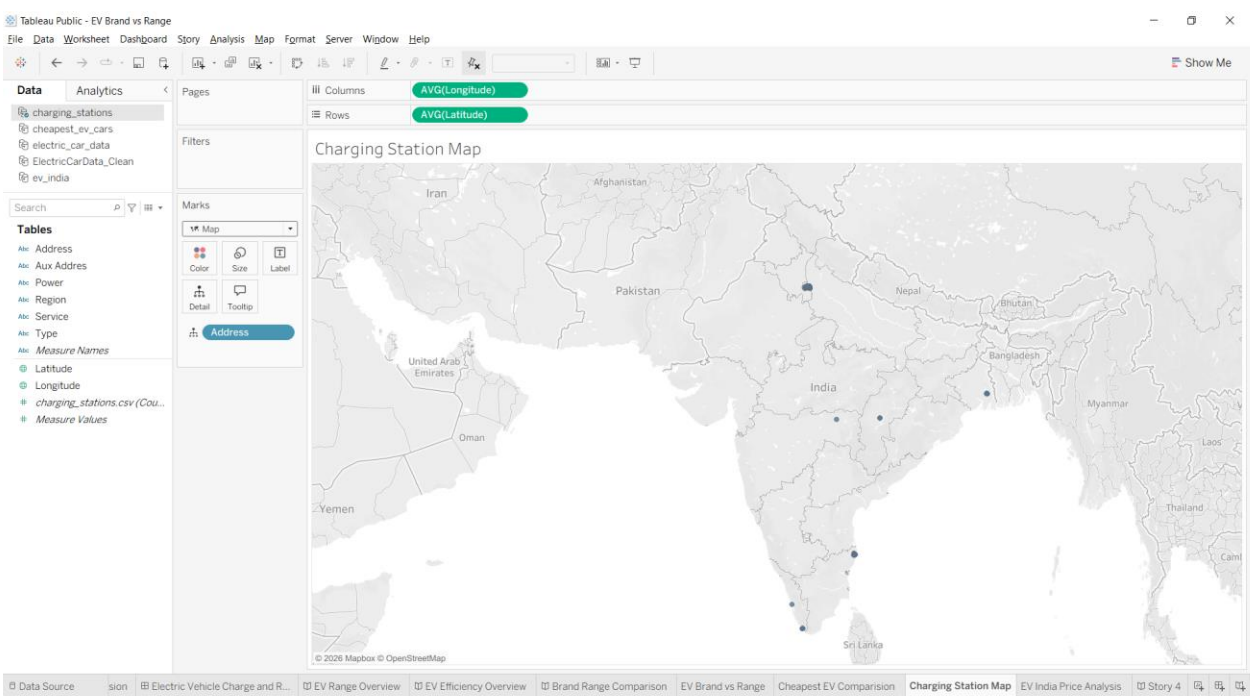


Figure 19.1: Charging Station Analysis Map in Tableau

CHAPTER 20

EV INDIA PRICE ANALYSIS

The **EV India Price Analysis** worksheet was created in Tableau to compare the prices of electric vehicles available in the Indian market. Price is one of the most significant factors influencing the purchasing decisions of customers, and this analysis provides a clear comparison of different EV models based on their cost.

The visualization enables users to identify affordable as well as premium electric vehicles and compare them with other available models. It helps in understanding pricing trends across different manufacturers and provides valuable insights into the Indian electric vehicle market.

Using Tableau's interactive visualization features, users can easily explore and compare vehicle prices, making the analysis more informative and user-friendly. This visualization supports better decision-making for customers, businesses, and researchers interested in the EV market.

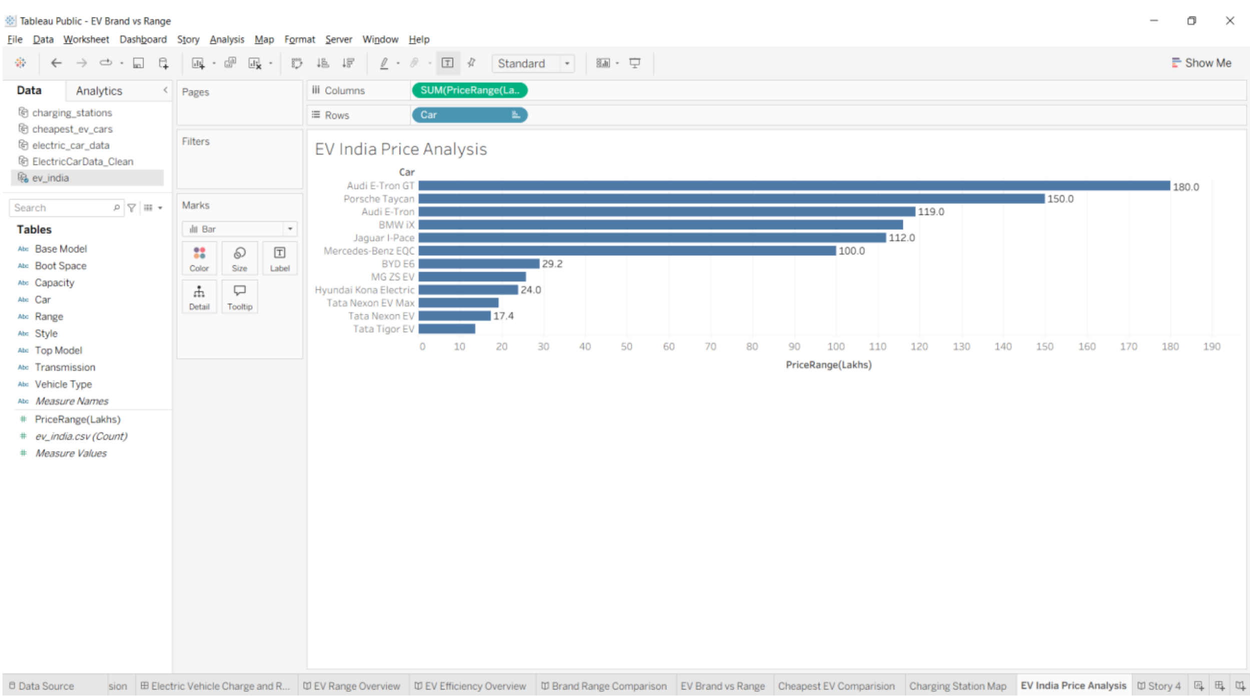


Figure 20.1: EV India Price Analysis Worksheet in Tableau

CHAPTER 21

DASHBOARD DEVELOPMENT

The final dashboard was developed in Tableau by combining multiple worksheets into a single interactive interface. The dashboard integrates important visualizations such as EV Range Analysis, EV Efficiency Analysis, Brand Comparison, Price Analysis, and Charging Station Analysis to provide a comprehensive view of the electric vehicle dataset.

Interactive features such as filters, highlighting, and dynamic charts enable users to explore the data efficiently. The dashboard allows users to compare different electric vehicles, analyze manufacturer performance, evaluate driving range, and understand pricing trends from a single screen.

The dashboard provides a user-friendly environment for analyzing electric vehicle data and supports faster decision-making by presenting multiple insights in one place. It demonstrates the effectiveness of Tableau as a business intelligence and data visualization tool.

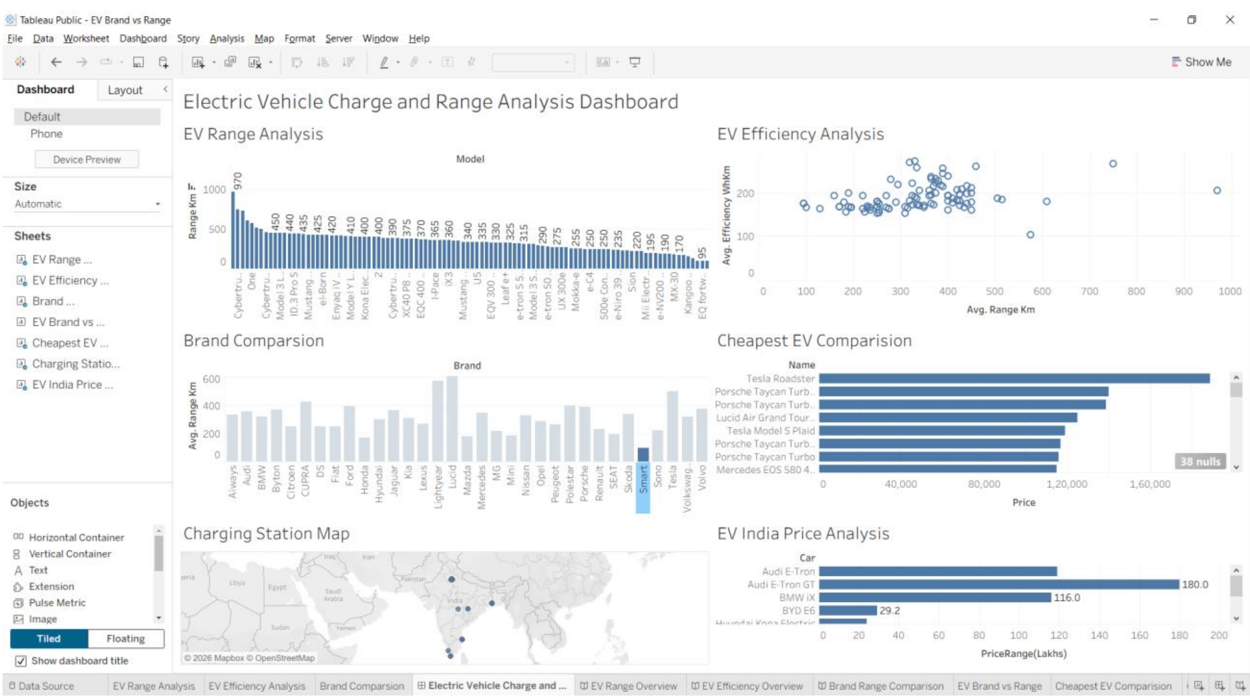


Figure 21.1: Interactive Dashboard Developed in Tableau

CHAPTER 22

STORY PRESENTATION

The Story feature in Tableau was used to present the complete analysis in a logical and sequential manner. A story consists of multiple visualizations arranged in an organized flow, enabling users to understand the analysis step by step.

The developed story combines important insights from EV Range Analysis, EV Efficiency Analysis, Brand Comparison, Price Analysis, and Charging Station Analysis into a single presentation. This approach makes it easier for users to interpret the results and understand the relationships between different aspects of the electric vehicle dataset.

The Tableau Story provides an effective way to communicate analytical findings and supports informed decision-making by presenting the results in a simple, interactive, and visually appealing format.

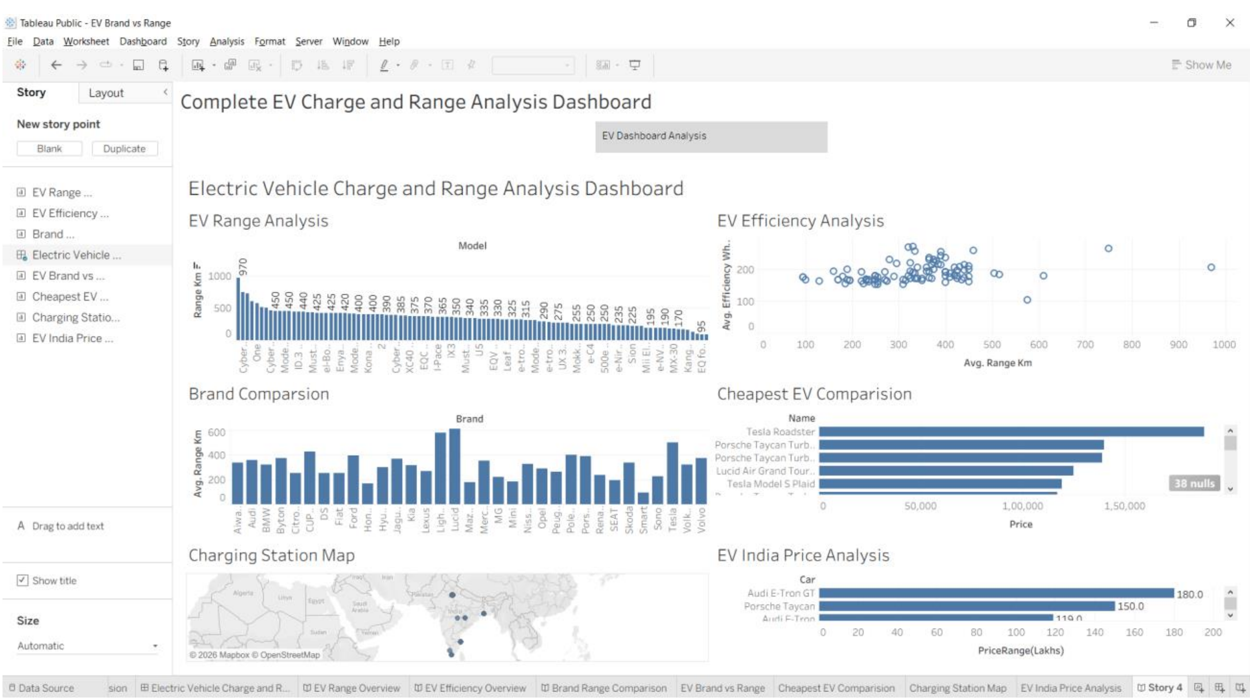


Figure 22.1: Tableau Story Presentation

CHAPTER 23

RESULTS AND DISCUSSION

The project was successfully completed by integrating MySQL and Tableau to analyze and visualize electric vehicle data. The dataset was imported into MySQL, where SQL queries were executed to validate, retrieve, and organize the required information. The processed data was then connected to Tableau for creating interactive visualizations.

Multiple worksheets, including EV Range Analysis, EV Efficiency Analysis, Brand Comparison, Cheapest EV Comparison, Charging Station Analysis, and EV India Price Analysis, were developed to explore different aspects of the dataset. These visualizations helped in identifying trends, comparing vehicle performance, and understanding pricing patterns.

An interactive dashboard and Tableau story were also created to present the complete analysis in a clear and user-friendly manner. The dashboard enabled users to explore multiple insights from a single interface, while the story presented the findings in a logical sequence.

Overall, the project demonstrated that combining MySQL with Tableau provides an effective solution for managing, analyzing, and visualizing electric vehicle data. The developed visualization tool simplifies data interpretation and supports better decision-making through meaningful graphical insights.

CHAPTER 24

CONCLUSION

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** was successfully developed using MySQL and Tableau. The project demonstrated how raw electric vehicle data can be organized, managed, and transformed into meaningful visualizations that are easy to understand.

MySQL was used to create and manage the database, while Tableau was used to build interactive worksheets, dashboards, and stories. These visualizations enabled effective analysis of electric vehicle range, efficiency, pricing, manufacturer comparison, and charging station information.

The developed system simplifies data analysis by presenting complex information in a graphical format, allowing users to identify trends and compare different electric vehicles efficiently. The interactive dashboard further enhances the user experience by providing multiple insights in a single view.

Overall, this project successfully achieved its objectives and demonstrated the importance of data visualization in supporting better understanding and decision-making in the field of electric vehicles.

CHAPTER 25

FUTURE SCOPE

The current project provides an effective platform for analyzing and visualizing electric vehicle data. However, there are several opportunities for future enhancements that can improve the functionality and usefulness of the system.

In the future, the project can be extended by integrating real-time electric vehicle data from online sources. Live updates on vehicle specifications, charging station availability, and market prices would make the analysis more accurate and useful.

Advanced analytical techniques such as machine learning and predictive analytics can also be incorporated to forecast electric vehicle demand, battery performance, and charging requirements. Additional visualization features and interactive dashboards can further improve the user experience.

The system can also be deployed as a web-based application, allowing users to access dashboards and reports from any device. These enhancements would make the project more scalable, interactive, and valuable for researchers, businesses, and electric vehicle users.

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Project Links

GitHub Repository:

https://github.com/karthikmanchala23-cell/EV-Charge-Range-Analysis

Tableau Public Dashboard:

https://public.tableau.com/app/profile/manchala.karthik/viz/Electric_Vehicle_Charge_and_Range_Analysis/ElectricVehicleChargeandRangeAnalysisDashboard?publish=yes